ECE524: Foundations of Reinforcement Learning **Spring 2024**

Lecture 2: Bellman Equations

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1 MDP Basics

At the very end of last lecture, we discussed that there are two types of policies—Markov policies and general policies. We also introduce two types of value functions— V-value and Q-value. We remark that the definitions of $V_h^\pi(s)$ and $Q_h^\pi(s,a)$ only apply to Markov policies for $h > 1$, as general policies can depend on the past history, thus the future cumulative reward at step h no longer only depends on the current state (or state-action pair) but also the past history. Nevertheless, the definitions of $V_1^\pi(s)$ and $Q_1^\pi(s,a)$ still apply to general policies as there is no past history.

General policies can be rather difficult to handle. Our following proposition says that in the context of MDP, we can always restrict our attention to Markov policies without loss of generality:

Proposition 1. *For any general policy π, there always exists a Markov policy μ such that $V_1^\pi(s_1) = V_1^\mu(s_1)$.*

That is, the total expected rewards for the above two policies will be the same. In particular, this proposition implies that, for the optimal policy $\pi^\star$ which can be possibly a general policy that achieves the maximum value, there will also be a Markov policy $\mu^\star$ that achieves the same value. Thus, we can conclude that for MDP, there will always be at least one optimal policy that is a Markov policy.

Proof of Proposition 1. By definition of value, we have:

$$V_1^\pi(s_1) = \sum_{h=1}^{H} \mathbb{E}_\pi[r_h(s_h,a_h)|s_1] = \sum_{h=1}^{H}\sum_{s_h,a_h} P_h^\pi(s_h,a_h|s_1) r_h(s_h,a_h) \tag{1}$$

where $P_h^\pi(s,a|s_1)$ (which is also known as the visitation measure) denotes the probability of visiting state-action pair (s,a) at step h if starting from initial state s_1 and following policy π. Since initial state s_1 is fixed, we will use succinct notation $P_h^\pi(s,a) = P_h^\pi(s,a|s_1)$ to hide the redundant dependency on a fixed quantity. Since on the RHS of (1), only the visitation measure depends on the policy, we know that in order to prove $V_1^\pi(s_1) = V_1^\mu(s_1)$, we only need to show:

$$P_h^\pi(s,a) = P_h^\mu(s,a) \text{ for all } (s,a,h) \tag{2}$$

For the rest of the proof, we will explicitly construct a Markov policy μ based on π, and show their visitation measures are equal. Our construction is based on a simple intuition that we want the Markov policy μ to be policy π averaging (or taking expectation) over the past history. In math, we denote the trajectory up to step h as $\tau_h = (s_1, a_1, \ldots, s_h)$, and define:

$$\mu_h(a|s) = \mathbb{E}_\pi[\pi_h(a|\tau_h)|s_h = s] = \begin{cases} \frac{\sum_{\tau_h: s_h = s} P_h^\pi(\tau_h)\pi_h(a|\tau_h)}{P_h^\pi(s_h = s)} & \text{if } P_h^\pi(s_h = s) \neq 0; \\ \text{arbitrary policy} & \text{otherwise.} \end{cases}$$

Now we show that our construction of Markov policy satisfies (2). We do so by induction.

For the base case, it is clear that $P_1^\pi(s_1) = P_1^\mu(s_1)$ as the probability to visit initial states in the first step does not depend on the policy.

For any fixed $h \in [H]$, we proceed with two steps:

1. show that $[\forall s, P_h^\pi(s) = P_h^\mu(s)] \Rightarrow [\forall (s,a), P_h^\pi(s,a) = P_h^\mu(s,a)]$
2. show that $[\forall (s,a), P_h^\pi(s,a) = P_h^\mu(s,a)] \Rightarrow [\forall s, P_{h+1}^\pi(s) = P_{h+1}^\mu(s)]$

For step 1, we note that:

$$P_h^\mu(s,a) = P_h^\mu(s)\mu_h(a|s) \overset{(a)}{=} P_h^\pi(s)\mu_h(a|s) \overset{(b)}{=} \sum_{\tau_h : s_h = s} P_h^\pi(\tau_h)\pi_h(a|s) = \sum_{\tau_h : s_h = s} P_h^\pi(\tau_h, a) = P_h^\pi(s,a)$$

where (a) follows from induction hypothesis, and (b) is due to our particular construction of policy μ_h. The remaining follows from basic probability calculations.

For step 2, we have:

$$\begin{aligned} P_{h+1}^\mu(s_{h+1}) &= \sum_{s_h, a_h} P_h^\mu(s_h, a_h, s_{h+1}) \overset{(c)}{=} \sum_{s_h, a_h} \mathbb{P}_h(s_{h+1}|s_h, a_h) P_h^\mu(s_h, a_h) \\ &\overset{(d)}{=} \sum_{s_h, a_h} \mathbb{P}_h(s_{h+1}|s_h, a_h) P_h^\pi(s_h, a_h) \overset{(e)}{=} \sum_{s_h, a_h} P_h^\pi(s_h, a_h, s_{h+1}) = P_{h+1}^\pi(s_{h+1}) \end{aligned}$$

where (d) follows from induction hypothesis. Both (c) and (e) are due to the key Markov property of MDP, where state transition $P_h^\mu(s_{h+1}|s_h, a_h) = P_h^\pi(s_{h+1}|s_h, a_h) = \mathbb{P}_h(s_{h+1}|s_h, a_h)$ which is independent of the policy and past history. This finishes the proof. $\square$

From now on, we will purely focus on Markov policies when discussing MDPs.

Notation. We will also introduce a few notations that will be commonly used in the class. We have introduced the values $V_h^\pi(s), Q_h^\pi(s,a) \in \mathbb{R}$ which are scalars. From time to time, we will be interested in the value function V_h^π, Q_h^π themselves without specifying the state or action. In the tabular setting which has a finite number of states and actions, we denote S as the number of states, and A as the number of actions. Then, value function V_h^π can be viewed as a vector in $\mathbb{R}^S$ where each coordinate corresponds to a particular state, and the value of the s^{th} coordinate is $V_h^\pi(s)$. Similarly, we can view Q_h^π as a vector in $\mathbb{R}^{SA}$ where each coordinate corresponds to a particular state-action pair. We will also frequently encounter the expectation of form $\mathbb{E}_{s' \sim \mathbb{P}(\cdot|s,a)} V(s')$ for certain value function V. We succinctly write such expectation using the notation $\mathbb{E}_{s' \sim \mathbb{P}_h(\cdot|s,a)} V(s') = (\mathbb{P}_h V)(s,a)$ to hide the dummy variable s'. You can understand this notation as matrix vector multiplication where transition matrix $\mathbb{P}_h \in \mathbb{R}^{SA \times S}$, and the resulting vector $(\mathbb{P}_h V) \in \mathbb{R}^{SA}$.

1.1 Policy evaluation

Recall the definition of value for Markov policies

$$\begin{aligned} V_h^\pi(s) =& \mathbb{E}_\pi[\sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'}) \mid s_h = s]. \\ Q_h^\pi(s,a) =& \mathbb{E}_\pi[\sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'}) \mid s_h = s, a_h = a]. \end{aligned}$$

Policy evaluation is the task which asks the following question: given transition $\mathbb{P}$, reward r and a fixed Markov policy π, how to compute the total value of this policy, which is $V_1^\pi(s_1)$. A naive approach is to directly compute using the definition above. By expanding the expectation (over the entire trajectory $\tau_H = (s_1, a_1, \ldots, s_H, a_H)$), we have:

$$V_1^\pi(s_1) = \sum_{\tau_H} \left[P^\pi(\tau_H|s_1) \sum_{h=1}^{H} r_h(s_h, a_h) \right] \tag{3}$$

where the probability $P^\pi(\tau_H|s_1) = \pi_1(a_1|s_1)\mathbb{P}_1(s_2|s_1, a_1)\pi_2(a_2|s_2)\mathbb{P}_2(s_3|s_2, a_2)\cdots$. While this approach can perform policy evaluation, it is highly inefficient—we note that the summand τ_H in (3) has $(SA)^H$ possibilities, which is exponentially large in horizon H. Therefore, the computation is rather prohibitive even for a reasonably large H.

Is there any way that we can evaluate the value significantly faster? The key idea lies in *dynamic programming*. One crucial observation is that the value functions at different steps satisfy the following recursive equations, which are also known as the **Bellman equation**:

$$\begin{cases} V_h^\pi(s) = \sum_{a\in\mathcal{A}} Q_h^\pi(s,a)\pi_h(a|s) \\ Q_h^\pi(s,a) = r_h(s,a) + (\mathbb{P}_h V_{h+1}^\pi)(s,a) \end{cases}$$

recall that by our notation convention $(\mathbb{P}_h V_{h+1}^\pi)(s,a) = \mathbb{E}_{s'\sim\mathbb{P}_h(\cdot|s,a)} V_{h+1}^\pi(s')$. The Bellman equation can be easily derived from the definitions of values. It also directly gives a dynamic programming approach to compute the value:

Algorithm 1 Policy Evaluation using the Bellman Equation

Initialize $V_{H+1}^\pi(s) = 0$ for all state s.
for step $h = H, H-1, \ldots, 1$ **do**
  $Q_h^\pi(s,a) \leftarrow r_h(s,a) + (\mathbb{P}_h V_{h+1}^\pi)(s,a)$ for all (s,a),
  $V_h^\pi(s) \leftarrow \sum_{a\in\mathcal{A}} Q_h^\pi(s,a)\pi_h(a|s)$ for all s.

That is, we start with $V_{H+1}^\pi \in R^S$ as the zero vector, and then use the Bellman equation to sequentially compute $Q_H \in \mathbb{R}^{SA}$, then $V_H \in \mathbb{R}^S$, then $Q_{H-1} \in \mathbb{R}^{SA}$, then $V_{H-1} \in \mathbb{R}^S$, $\cdots$. [Insert a better picture of DP]. The overall computation is polynomial in S, A, H. In fact, the computation is only linear in H, and is thus much faster than the naive approach.

1.2 Optimal policy

After we learned how to compute the value for a fixed policy, the natural next step is to compute the optimal policy. Following the similar idea from previous subsection, we believe that dynamic programming is the key idea for computing many important quantities for MDPs, and we would like to also use dynamic programming to compute the optimal policy.

One straightforward guess is to modify the Bellman equation. We note only the first line of the Bellman equation explicitly depends on the policy π, which takes expectation over actions according to policy π. We expect the optimal policy to take actions that always maximize the Q-value. This gives rise to the **Bellman optimality equation**:

$$\begin{cases} V_h^\star(s) = \max_{a\in\mathcal{A}} Q_h^\star(s,a) \\ Q_h^\star(s,a) = r_h(s,a) + (\mathbb{P}_h V_{h+1}^\star)(s,a) \end{cases}$$

The Bellman optimality equation also gives a particular policy of taking action:

$$\pi_h^\star(a|s) = \mathbb{I}[a = \arg\max_{a'\in\mathcal{A}} Q_h^\star(s,a')] \tag{4}$$

It is not hard to see the Bellman equation of $\pi^\star$ coincides with the Bellman optimality equation. Therefore, $V^\star, Q^\star$ are also the value functions for policy $\pi^\star$. That is, $V_h^\star(s) = V_h^{\pi^\star}(s)$ and $Q_h^\star(s,a) = Q_h^{\pi^\star}(s,a)$ for all (s,a,h).

So far, we have guessed the Bellman optimality equation and defined the corresponding policy $\pi^\star$, it remains to show that $\pi^\star$ constructed this way is in fact an optimal policy.

Proposition 2. *The policy $\pi^\star$ defined in* (4) *is an optimal policy.*

Proof. We again use induction to prove this claim. In fact, we will prove an even stronger claim—we will show that

$$\forall(s,a,h), \qquad V_h^\star(s) = \max_\pi V_h^\pi(s), \qquad Q_h^\star(s,a) = \max_\pi Q_h^\pi(s,a) \tag{5}$$

where the maximization is taken over all Markov policies. That is, $\pi^\star$ is not only the optimal policy that achieves the highest value for $V_1^\pi(s_1)$, but also achieves the highest value starting from all states (or state-action pairs) at every step simultaneously!

For the base case, it is clear that for all state s, $V_{H+1}^\star(s) = \max_\pi V_{H+1}^\pi(s) = 0$ since there is no reward to collect after step H.

For any fixed $h \in [H]$, we proceed with two steps:

1. show that $[\forall s, V_{h+1}^\star(s) = \max_\pi V_{h+1}^\pi(s)] \Rightarrow [\forall(s,a), Q_h^\star(s,a) = \max_\pi Q_h^\pi(s,a)]$
2. show that $[\forall(s,a), Q_h^\star(s,a) = \max_\pi Q_h^\pi(s,a)] \Rightarrow [\forall s, V_h^\star(s) = \max_\pi V_h^\pi(s)]$

For step 1, we have:

$$\begin{aligned}\max_\pi Q_h^\pi(s,a) &\overset{(a)}{=} r_h(s,a) + \max_\pi \mathbb{E}_{s'\sim\mathbb{P}_h(\cdot|s,a)} V_{h+1}^\pi(s') = r_h(s,a) + \max_\pi \sum_{s'} \mathbb{P}_h(s'|s,a) V_{h+1}^\pi(s') \\ &\overset{(b)}{=} r_h(s,a) + \sum_{s'} \mathbb{P}_h(s'|s,a) \max_\pi V_{h+1}^\pi(s') \overset{(c)}{=} r_h(s,a) + \sum_{s'} \mathbb{P}_h(s'|s,a) V_{h+1}^\star(s') \overset{(d)}{=} Q_h^\star(s,a)\end{aligned}$$

where (a) and (d) are by the Bellman equation and the Bellman optimality equation respectively. (c) is due to the induction hypothesis. The key step here is (b)—in general, we can not move maximization inside summation as $\max_x(f(x)+g(x)) \le \max_x f(x) + \max_x g(x)$. However, if we know the maximization of each individual function is achieved at same $x^\star$ simultaneously, then we have $\max_x(f(x)+g(x)) = \max_x f(x) + \max_x g(x)$. This is exactly what happens in (b) since our induction hypothesis guarantees that $\max_\pi V_{h+1}^\pi(s')$ is achieved at $\pi^\star$ for all state s' simultaneously.

Similarly, for step 2, we have:

$$\begin{aligned}\max_\pi V_h^\pi(s) &\overset{(a)}{=} \max_\pi \sum_a Q_h^\pi(s,a)\pi_h(a|s) = \max_{\pi_h} \max_{\pi_{h+1:H}} \sum_a Q_h^\pi(s,a)\pi_h(a|s) \overset{(b)}{=} \max_{\pi_h} \sum_a (\max_{\pi_{h+1:H}} Q_h^\pi(s,a))\pi_h(a|s) \\ &\overset{(c)}{=} \max_{\pi_h} \sum_a Q_h^\star(s,a)\pi_h(a|s) \overset{(d)}{=} \max_a Q_h^\star(s,a) \overset{(e)}{=} V_h^\star(s)\end{aligned}$$

Again (a) and (e) are by the Bellman equation and the Bellman optimality equation respectively. (b) is because $Q_h^\pi(s,a)$ only depends on the policy starting from step $h+1$ (since the action at step h is already specified to be a, thus does not depend on the policy at step h), and the same reasoning indicates the maximization is achieved at $\pi^\star$ simultaneously for all (s,a) pair by induction hypothesis. (c) is also by the induction hypothesis. (d) is because the maximizing policy is achieved when putting probability 1 on the action that achieves the highest Q-value. This finishes the proof. □